Portfolio exposure across two Qwen3-8B configurations

Matched re-evaluation of 100 portfolio questions

The original configuration returned 35/100 correct final totals. The reasoning-enabled configuration returned 98/100 on the same prompts. The accuracy gap between the two controlled conditions changed from 83.3 to 0.0 percentage points.

Task and scoring outcome	Original	Reasoning enabled
Correct total with identical product allocations	24/24	24/24
Correct total with different product allocations	4/24	24/24
Correct direct total from published tables	2/26	25/26
Correct total reported alongside the ledger	5/26	25/26
All reported ledger fields within tolerance	23/26	25/26
Calculator correct using the reported ledger	23/26	25/26
Ledger correct but reported total wrong	18/26	0/26

Each entry counts prompts meeting the stated criterion. A correct total lies within 0.5 percentage points of the reference value. The last three rows describe the same 26 ledger prompts. A calculator weights each reported allocation by its reported investment amount. Both remaining responses reached 4,096 tokens with an empty final answer. They used the fragmented MyMap 4/MyMap 5 account, once per response format.

What the controlled questions measure. In one family, all four products allocate 55% to Asset A. Omitting a lot still leaves a weighted average of 55%. The paired condition gives products different allocations, so errors in the relative amounts can change the answer. The recheck covers both conditions for each of 24 families, using their merged, direct-answer prompts.

Evidence from published tables. The other 52 prompts use cleaned holdings tables for eight funds from Vanguard and BlackRock. Thirteen constructed accounts each appear in two presentations and two response formats. The original configuration reported correct ledger fields alongside a wrong total in 18 cases. Recomputing those fields gave the correct exposure in all 18. Under the second configuration, all 25 returned ledgers had correct fields and a correct total.

Reproduction and scope. Both runs use the same Qwen3-8B Q4_K_M weights, input text and JSON-object constraint. The original uses greedy decoding with a 512-token limit. The second enables reasoning, uses temperature 0.6 and allows 4,096 tokens. Sampling, seed and runtime settings also differ, so the comparison estimates the combined configuration change on previously examined cases. Strict scoring includes truncated and invalid responses in the denominator. The final 31 prompts ran on a replacement CPU host after the model and inference-program hashes were verified.

The companion archive includes `data/protocol.json`, `results/analysis.json` and `results/evaluation/raw.jsonl`.